

4 Method

This section describes the NITSCI model (Fig. 2), its training objective, the two-stage training procedure and the decision rule. Given an image $x \in \mathbb{R}^{3 \times 448 \times 448}$, a backbone produces a feature map that is turned into N tokens. Two experts process the tokens, exchange information, and are pooled and fused by a gate; two heads produce a class distribution and a continuous grade.

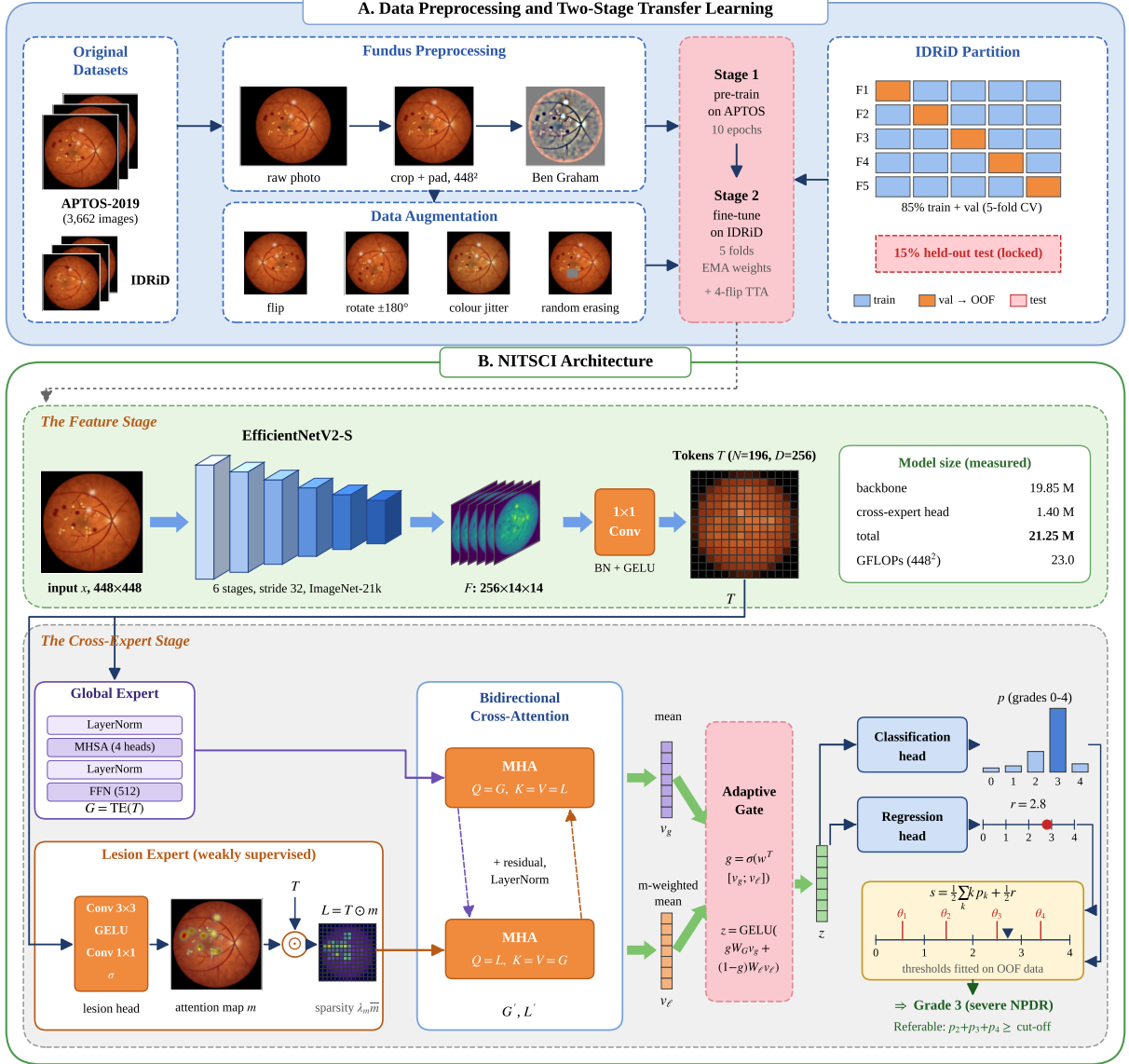


Figure 2: Overview of the proposed method. (A) Data preparation and two-stage transfer learning: APTOS-2019 and IDRiD images are pre-processed identically; the full model is pretrained on APTOS-2019 (stage 1) and fine-tuned on five stratified IDRiD folds (stage 2), while a stratified 15% IDRiD test set is locked away; out-of-fold predictions fix the decision rule, the grade thresholds $\theta_1, \dots, \theta_4$ and the referral cut-off. (B) NITSCI architecture: EfficientNetV2-S features are projected to $N = 196$ tokens, processed by a global expert and by a lesion expert $L = T \odot m$ built from a learned lesion map m , coupled by bidirectional cross-attention, pooled and fused by an adaptive gate g ; classification and regression heads are combined into the grade score s , whose thresholds come from panel A. Fundus photographs, feature maps and the probability values shown are synthetic illustrations, not patient data or model outputs.

4.1 Backbone and tokenisation

We use EfficientNetV2-S [30] pretrained on ImageNet-21k and fine-tuned on ImageNet-1k [56], taken from the `timm` library [67]. Its last stage (stride 32, 256 channels) gives a 14×14 map for a 448×448 input. A 1×1 convolution with batch normalisation and GELU projects it to $F \in \mathbb{R}^{D \times h \times w}$ with $D = 256$, which is flattened into $N = hw = 196$ tokens $T \in \mathbb{R}^{N \times D}$.

4.2 Lesion-attention map and lesion expert

A light head (3×3 convolution to $D/4$ channels, GELU, 1×1 convolution to one channel, sigmoid) predicts a spatial map $m \in [0, 1]^{h \times w}$. No lesion annotation is used: m is learned end-to-end from grade labels and is encouraged to be sparse by an ℓ_1 penalty (Eq. 7). The lesion expert is the re-weighted token set

$$L = T \odot m, \quad L_i = m_i T_i, \quad i = 1, \dots, N. \quad (2)$$

4.3 Global expert

The global expert is one pre-norm Transformer encoder layer [45] with four heads, a feed-forward width of $2D$ and dropout 0.1, applied to all tokens: $G = \text{TE}(T)$. It models context across the whole retina, such as the spatial spread of lesions and their relation to the optic disc and macula.

4.4 Bidirectional cross-attention

Each expert queries the other with multi-head attention (four heads, dropout 0.1):

$$Z_g = \text{MHA}(Q=G, K=L, V=L), \quad G' = \text{LN}(G + Z_g), \quad (3)$$

$$Z_\ell = \text{MHA}(Q=L, K=G, V=G), \quad L' = \text{LN}(L + Z_\ell). \quad (4)$$

The global stream thus gathers lesion evidence, and the lesion stream receives the context needed to interpret it.

4.5 Pooling and adaptive gate

The global stream is mean-pooled, $v_g = \frac{1}{N} \sum_i G'_i$, and the lesion stream is pooled with the lesion weights, $v_\ell = \sum_i m_i L'_i / (\sum_i m_i + \epsilon)$. A scalar gate decides, per image, how much each expert contributes:

$$g = \sigma(w^\top [v_g; v_\ell] + b), \quad z = \text{GELU}(g W_G v_g + (1 - g) W_L v_\ell). \quad (5)$$

This is a two-expert soft mixture [22]; g is also reported for interpretability.

4.6 Dual head and grade score

Two heads share z : a classification head (LayerNorm, dropout 0.3, linear) gives logits and probabilities $p \in \Delta^4$, and a regression head of the same form gives $r = f_r(z) + 2$, centred on the middle grade. Their outputs are combined into a continuous score

$$s = \frac{1}{2} \sum_{k=0}^4 k p_k + \frac{1}{2} r. \quad (6)$$

A grade is obtained by comparing s with four ordered thresholds $\theta_1 < \dots < \theta_4$: $\hat{y} = \#\{j : s \geq \theta_j\}$.

4.7 Training objective

For a mini-batch the loss is

$$\mathcal{L} = \text{CE}_{w,\varepsilon}(p, y) + \lambda_r \text{SmoothL1}(r, y) + \lambda_m \overline{m}, \quad (7)$$

where $\text{CE}_{w,\varepsilon}$ is cross-entropy with label smoothing $\varepsilon = 0.05$ [53] and class weights $w_c \propto n_c^{-1/2}$ (normalised to mean one, computed on the training fold), $\overline{m}$ is the mean of the up-sampled lesion map, $\lambda_r = 0.5$ and $\lambda_m = 0.01$. The square-root weighting is a compromise between ignoring and fully inverting the imbalance.

4.8 Two-stage training

Stage 1 (APTOS). The full model, initialised from ImageNet-21k backbone weights, is trained on APTOS-2019 for 10 epochs (learning rate 3×10^{-4} , EMA decay 0.999). **Stage 2 (IDRiD).** For each of the five folds, a model initialised from the stage-1 weights is fine-tuned for at most 25 epochs (learning rate 2×10^{-4} , EMA decay 0.99, early stopping with patience 8). In both stages we use AdamW [68] with weight decay 10^{-4} , a backbone learning rate one third of the head learning rate, a one-cycle cosine schedule with 10% warm-up [69], mixed precision [70], gradient clipping at norm 2, and an exponential moving average (EMA) of weights [59] that is used for validation and saved. The checkpoint with the highest sum of validation accuracy and quadratic weighted kappa (fixed thresholds 0.5, 1.5, 2.5, 3.5, no TTA) is kept.

4.9 Inference and decision rule

Each fold model predicts every test image under four views (identity, horizontal flip, vertical flip, both); probabilities and regression outputs are averaged over views and over the five folds (20 predictions per image). The decision rule is chosen *on out-of-fold (OOF) predictions only* among three candidates: (i) argmax of p ; (ii) rounding s with thresholds 0.5, 1.5, 2.5, 3.5; (iii) optimised thresholds, found by coordinate search (four rounds, 80 candidate values per threshold, constrained to remain ordered) maximising OOF accuracy. The chosen rule was **[rule]**, with thresholds (**[thr]**).

Referable DR. For the screening task we use $P(\text{referable}) = p_2 + p_3 + p_4$ and select its operating threshold on OOF predictions by maximising Youden’s J [71] (selected value **[refthr]**).

4.10 Model size and cost

Table 2 lists the parameter counts and floating-point operations, measured with PyTorch [72] on a single 448×448 input. The cross-expert head adds 7% to the parameters of the backbone.

Table 2: Model size and cost for one 448×448 image (FLOPs counted as $2 \times$ multiply-accumulates).

Component	Parameters (M)	Notes
EfficientNetV2-S feature extractor	19.85	256 output channels
Cross-expert head (all modules after the backbone)	1.40	$D = 256$, 4 heads
NITSCI total	21.25	23.0 GFLOPs
Plain baseline (same backbone, GAP, same dual head)	19.92	–
Inference latency per image (fp16, batch 16, [gpu])	[mspi] ms per model and view	